

Tower of Hanoi: Solution

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1. 7

Move the smallest disc onto either the first or second peg, and place the middle one onto the remaining peg. For example's sake, place the smallest on the third peg and the middle one on the second peg. This is 2 moves so far.

Then, place the small disc on the medium disc. This allows the large disc to be moved to the third peg. This is another 2 moves.

Place the small disc onto the first peg. The medium disc can then be placed on the large disc, and the small disc can finally be placed on top. These are the last 3 moves.

$$2 + 2 + 3 = \boxed{7}$$

2. $H(4, 3) = 15$; $H(5, 3) = 31$; $H(10, 3) = 1023$

In order to not make the solution too lengthy (as would be the case for $H(10, 3)$), consider the middle case, $H(5, 3)$. Instead of starting the solution all over again, we can consider what would happen if $H(4, 3)$ was already completed and we simply added another disc. If had a tower of 4 discs on the second peg, and we placed our new disc on the first peg, we can find how many moves it would take to solve this.

For brevity, the sequence will be written as xT , where x is the number of the disc that is being moved, with 1 being the shortest and 5 being the longest, and where T is the peg to which that x is being moved, denoted by L, M, R , or left, middle, right.

5R, 1L, 2R, 1R, 3L, 1M, 2L, 1L 4R, 1R, 2M, 1M, 3R, 1L, 2R, 1R.

$$H(4, 3) + 16 = 31$$

If we wanted to solve $H(4, 3)$, we could take the solution to $H(3, 3)$ and add the additional 8 moves.

An even more concise way is to consider that the only moves you have to take are solving $H(n - 1, k)$, moving the new disc, and then resolving $H(n - 1, k)$ on the new peg.

3. $2^n - 1$

A recursive argument can be made. Looking at the previous question, we see that $H(n+1, 3) = 2H(n, 3) + 1$. We can prove this recursive argument and get a general formula using both

logic and induction.

From observation, it appears that $H(n, 3) = 2^n - 1$. We have already proved this for the base case of $n = 3$, and for $n = 1$, we get $2^1 - 1 = 1$, which must clearly be true.

Let $H(n, 3) = 2^n - 1$ be true for some $i > 0$. Let $S_n = H(n, 3)$. Then, with $i + 1$ discs, we know that the first set of moves would be to move all the i discs off of the $(i + 1)^{\text{th}}$ disc. Then, move the $(i + 1)^{\text{th}}$ disc onto another peg, and finally put all i remaining discs from the stack back on top of that disc. This results in a total of $2S_i + 1$ moves.

$$S_{i+1} = 2S_i + 1 = 2(2^i - 1) + 1 = 2^{i+1} - 1$$

It can also be shown that for any disc k , where k is the disc's original position from bottom to top (ex: the largest disc = n), it takes $2^k - 1$ moves to transfer it to completion.

$$\sum_{k=1}^n 2^{k-1} = 2^n - 1$$

Both methods demonstrate that for some positive integer n , $H(n, 3) = \boxed{2^n - 1}$

4. (a) $H(1, 1), H(2, 2)$

It is impossible to move one disc onto another peg if there is only one peg. Similarly, if there are two discs and two pegs, the smaller disc can be moved onto the second peg, but nothing else can be done.

Similarly, it should make sense that for $H(n, 2)$, if $n \geq 2$, there are no solutions.

- (b) $H(4, 4) = 9; H(5, 5) = 11; H(10, 10) = 21$

Consider the middle question, $H(5, 5)$. We know that it will take 4 moves to move the top 4 discs onto their own pegs. However, we still need to move disc 5 onto its own peg. We can move one of the smaller discs on top of the larger discs, making room for disc 5 to go on another peg. We then need to take the small disc and place it onto the peg that disc 5 just left. This is an additional 3 moves. Redoing the puzzle then takes another 4 moves.

$$4 + 4 + 3 = 11$$

- (c) $2x + 1$

Using the logic from the previous question,

$$H(x, x) = (x - 1) + 3 + (x - 1)$$

$$H(x, x) = \boxed{2x + 1}$$

5. It is not possible to solve the Tower of Hanoi in an even number of moves, even in a nonoptimal setting.

We see that in an optimal setting, the number of moves it takes to complete the game is $2^n - 1$, which will always be an odd number for any positive integer n . Taking a nonoptimal number of moves to complete the Tower of Hanoi simply means making a mistake in placing a disc where it does not belong; this will always be n moves of mistakes and another n moves of fixing this mistake (it does not matter when this mistake is fixed). We have also strictly defined a move to be a transfer of a disc, so picking a disc up and placing it back down does not count as a move. Summing the optimal number of moves and the number of moves correlating to mistakes results in an odd integer.

6. (a) $g(3) = 5, g(5) = 9, g(15) = 29$

The logic is similar to question 4. Consider $g(5)$. We can move the top 4 discs onto their own pegs, then move 5 onto its own peg, and resolve the other 4 discs.

$$4 + 1 + 4 = 9$$

- (b) $n + 1$

We can see that the problem that we had in 4 is eliminated if we were to add an additional peg.

- (c) $2n - 1$

We can consider the optimal way to play the game for any k such that $g(n) = H(n, k)$, but since this will also hold true for all integers greater than or equal to k , we can think about this problem with an infinite number of pegs. In a situation like this, it would be best to place every piece out on its own peg so that no backtracking has to be taken, similar to (b).

$n - 1$ moves will need to be taken to get each of the $n - 1$ discs onto its own peg. The bottom disc can then be moved onto any other available peg, which takes precisely 1 move. Finally, since each disc is on its own peg, moving every other disc onto the bottom disc only takes another $n - 1$ moves.

$$(n - 1) + 1 + (n - 1) = \boxed{2n - 1}$$

7. (a) $S(2) = 4, S(3) = 13, S(5) = 121$

Consider the simplest case, $S(2)$. Let the orientation of the pegs be black-white-white (this will take the same number of moves as black-white-black).

Place the top disc onto another peg. In this solution, it will be placed in the middle peg. Then, move the larger disc into the last position. Finally, the orientation of the middle disc must be swapped, as it is currently white on top and flipping it would break the rules if it were to be placed on top of the larger disc. So, to re-orient it, flip it onto the black peg, and then transfer it onto the larger one. This results in 4 moves.

(b) $\frac{3^n - 1}{2}$

For every disc, the number of moves it takes to get it to its optimal position is 3^{k-1} .

$$\sum_{k=1}^n 3^{k-1} = \boxed{\frac{3^n - 1}{2}}$$

For additional information, such as a general form for $H(n, k)$, check out the Frame-Stewart Algorithm. It has not been proven optimal, though it is generally considered to be.